ECN 712 EE Schlee Fall 2025
Problem Set IX, Due FRIDAY, November 14 (by midnight)

1. Let $G = (N, \{S_i\}_{i \in N}, \{u_i\}_{i \in N})$ be a strategic-form game. Prove that $NE(G) \subseteq IDSDS(G)$: if a strategy profile is a Nash equilibrium, then it survives iterated deletion of strictly dominated strategies.

2. Find all Nash equilibria for *mixed extension* of the following Battle of the Sexes game with $b > a > 0$.

	Wrestling	Opera
Wrestling	b a	0 0
Opera	0 0	a b

3. Two firms are trying to develop a product. The firms simultaneously choose how much to spend on development, which can be any number in [0,1]. *Any amount spent is sunk.* The firm spending the most develops the product and earns *gross* profit of 1 from selling it (the other earns gross profit of zero); if they spend the same amount, then each earns a gross profit of 1/2. The firms are risk neutral.

   (a) Write down firm i's profit as a function of the strategy profile.

   (b) Does this game have a (pure-strategy) Nash equilibrium? Which assumptions of the Debreu-Fan-Glicksberg existence theorem does the game satisfy? Which does it not satisfy?

   (c) Consider the mixed extension of this game. Confirm that each firm using the uniform distribution on $[0, 1]$ is a Nash equilibrium of the mixed extension. Explain how would you show this if you weren't told the equilibrium in advance. (Assume that each firm's strategy is a continuous density function.) Does this game satisfy the assumption of Nash's (1950) existence theorem for mixed extensions?

4. MWG, Exercise 12.C.7.

5. MWG, Exercise 12.C.9. (Scholars and regulators often use the Herfindahl index to measure market concentration. In the context of the Exercise, that use is justifiable. The assumption that demand is linear can be dispensed with; the assumption of constant returns cannot.)[1]

6. *Cournot duopoly: testable implications.*[2] Consider a family of Cournot duopoly game $\{G_C(t)\}_{t \in \{0,1\}}$ in which, for $t = 0, 1$, each firm i has a constant-returns technology with unit cost of $k_i > 0$ and the inverse demand is $P(\cdot, t)$. For $t = 0, 1$, the inverse demand $P(\cdot, t)$ satisfies the usual properties: $P(0, t) > \max\{k_1, k_2\}$, $P(\cdot, t)$ is continuous and nonincreasing on $\mathbb{R}_+$, $P(q, t) < \min\{k_1, k_2\}$ for some $\bar{q} > 0$, and $P(\cdot, t)$ is differentiable on $[0, \bar{q}]$. For $t = 0, 1$, let $(q_1^t, q_2^t) \in NE(G_C(t))$. Suppose that $(q_2^0, q_2^1) \gg 0$ and that

$$\frac{q_1^1}{q_2^1} > \frac{q_1^0}{q_2^0} > 1. \tag{1}$$

   (a) Summarize the inequalities (1) in words.

   (b) For $t = 0, 1$, let $p^t = P(q_1^t + q_2^t, t)$, the price associated with the Nash equilibrium $q^t = (q_1^t, q_2^t)$ for the Cournot game indexed by demand parameter t. What if anything can be said about whether $p^1 \gtreqless p^0$? Prove it and summarize in words any testable implications regarding prices that you find.[3]

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[1] Is the Herfindahl index a convex function? What happens to it if the distribution of market shares becomes riskier (in the sense of the notes on increases in risk)? Is this a desirable property of a measure of concentration?

[2] This problem is inspired by Carvajal, A., R. Deb, J. Fenske, and J. Quah, "Revealed preference tests of the Cournot model," *Econometrica*, 81 (2013): 2351-2379.

[3] Note that I make *no* assumptions directly about about t enters $P(q, t)$; indeed the two demands could be the same, either could be uniformly higher than the other, or they could cross many times.